

PUCCH Format 0

1 PUCCH0 Processing Overview

Figure 1 shows the PUCCH0 receiver processing pipeline and the numbered interfaces

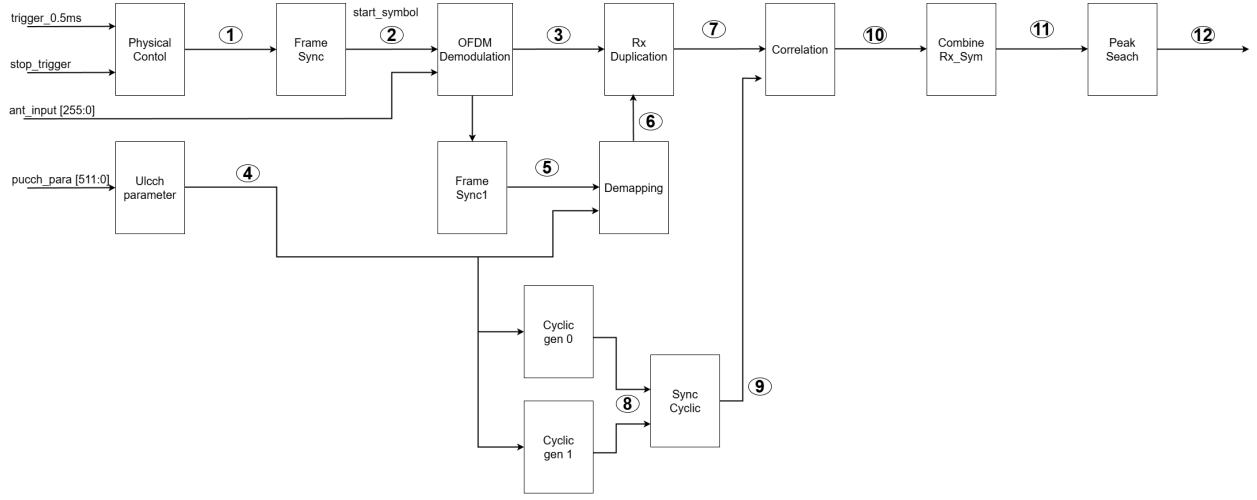


Figure 1: PUCCH0 processing.

For detailed input and output signals of each IP, see Fig. 2, which illustrates the PUCCH0 receiver processing pipeline in Vivado.

2 Interface Description

The numbered interfaces in Fig. 1 indicate the main functional signals and processing boundaries in the PUCCH0 receiver.

- **Interface 1 (Trigger processing):** Receives the **0.5 ms slot trigger** and generates **symbol triggers** (14 symbols per slot = 0.5 ms).
- **Interface 2 (CP/start generation):** Generates the **cyclic prefix (CP)** and **start-of-symbol** timing for the **OFDM demodulation** block.
- **Interface 3 (Resource-set output):** Outputs the selected **resource sets in the frequency domain**, with samples represented as **15-bit I** and **15-bit Q**.
- **Interface 4 (FAPI parameter parsing):** Parses the **PUCCH parameters** from **FAPI**.

- **Interface 5 (Frame timing generation):** Generates **frame timing information** (slot, symbol, RB) so the **demapping** block knows where the PUCCH data is located after OFDM demodulation.
- **Interface 6 (Data capture):** Captures the **useful (valid) data** for further processing.
- **Interface 7 (Resource duplication):** Duplicates the extracted resource **12 times** for multiple with cyclic.
- **Interface 8 (Cyclic sequence generation):** Generates **cyclic (pseudo) sequences**, with an interface of **16-bit I** and **16-bit Q**.
- **Interface 9 (Cyclic synchronization):** Aligns the cyclic sequences such that **sequence 0** comes first and **sequence 1** follows, corresponding to **symbol 0** and **symbol 1**, respectively.
- **Interface 10 (Correlation peaks):** After correlation, outputs the **peak values/metrics**.
- **Interface 11 (Peak combining):** Combines peak metrics across **multiple RX antennas** and across the **two symbols**.
- **Interface 12 (Peak search and ACK decision):** Finds the **maximum peak position/value** and determines the corresponding **ACK decision (decoded ACK bits)**.

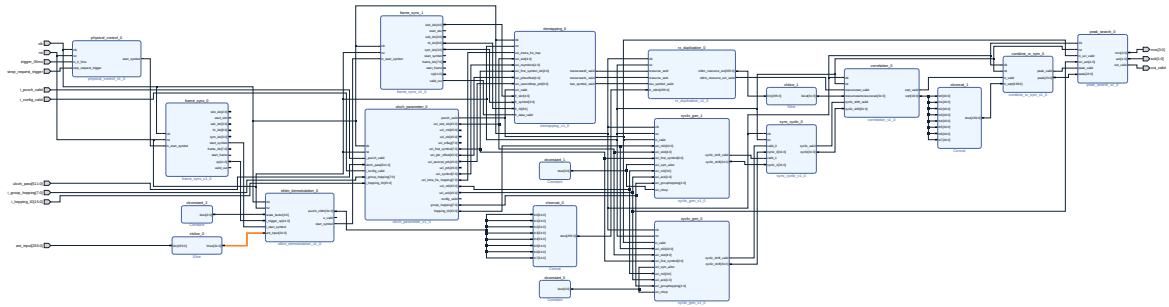


Figure 2: PUCCH0 processing on Vivado.